CHANDRACHUD PATI

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RESEARCH INTERESTS

My work is on **reliable quantum computing**: making results from today's noisy quantum machines trustworthy, from how a circuit is compiled to how a quantum job runs inside an HPC workflow. Going ahead, I want to carry these questions into **fault-tolerant quantum computing**: finding which layer of a logical computation loses accuracy, and meeting a precision target at the lowest total cost in code distance, magic states, and measurements.

EDUCATION

National Institute of Technology Rourkela · Rourkela, India *2019–2024*
Integrated M.Sc. in Mathematics · CGPA 7.72/10

Selected coursework: Probability and Statistics, Mathematical Methods, Numerical Analysis, Linear Programming, Statistics Laboratory, Basic Programming, and Computing Laboratory.

PUBLICATIONS AND MANUSCRIPTS

[1] **Per-Phase Fidelity Attribution for Quantum Compilers using HBR Decomposition**
Chandrachud Pati and Yogesh Simmhan. *IEEE Quantum Week (QCE) 2026. arXiv:2605.07876.*

[2] **Representation-Aware Portability of Quantum Circuits Across SDK Boundaries**
Chandrachud Pati, Abdur Rahman Hatim, and Yogesh Simmhan. *33rd IEEE International Conference on High Performance Computing, Data & Analytics (HiPC) 2026.*

[3] **SemQBench: Benchmarking Decomposition-Aware Runtime Policies for Quantum-HPC Search**
Chandrachud Pati and Yogesh Simmhan. *2nd SFWM Workshop, co-located with SC 2026. arXiv:2610.22886.*

[4] **Title withheld for double-blind review**
Chandrachud Pati et al. *Under review, IEEE IPDPS 2027. Part of ORNL's openQSE demo at SC26.*

[5] **Title withheld for double-blind review**
Chandrachud Pati et al. *Under review, IEEE IPDPS 2027.*

[6] **Title withheld for double-blind review**
Chandrachud Pati et al. *Under review, ACM SIGMETRICS 2027.*

POSTERS

[P1] **Schedulable but Infeasible: Pre-dispatch Checks for Quantum HPC Workflows**
Chandrachud Pati and Yogesh Simmhan. *IEEE HiPC 2026. Interactive demo.*

[P2] **Towards Forecasting Calibration Windows for Qubit Placement on IBM Quantum Systems**
Sakshi Chhabra*, **Chandrachud Pati***, and Yogesh Simmhan (*equal contribution). *IEEE HiPC 2026.*

[P3] **Pre-Execution QPU Time Estimation for Quantum Jobs in HPC Systems**
Aryan Singh Sisodiya, **Chandrachud Pati**, and Yogesh Simmhan. *IEEE HiPC 2026.*

RESEARCH EXPERIENCE

Indian Institute of Science (IISc) · Bengaluru, India *Feb 2026–Present*
Pre-doctoral Researcher, DREAM Lab · Advisor: Prof. Yogesh Simmhan

- **National Quantum Mission (NQM):** Building software for India's reference stack that runs quantum jobs inside cloud and HPC workflows.
- **openQSE:** Middleware for portable quantum–HPC workflows [4], part of ORNL's openQSE demo at SC26.
- **Reliable quantum computing:** First-authored work on where compilers lose quality [1, 2], how many measurements an answer needs [6, P1], and quantum–HPC runtimes [3, 5]; worked with two postgraduate students on their HiPC 2026 posters [P2, P3].

National Institute of Technology Rourkela · Rourkela, India *Sep 2023–Jan 2024*
Research Project – Hybrid Classical–Quantum Machine Learning · Advisor: Prof. Snehashish Chakraverty

- Built a hybrid quantum–classical neural network in Qiskit and PennyLane for highly imbalanced credit-card fraud detection.
- Found that how the circuits were compiled changed the model's accuracy, which led to my later compiler work.

INDUSTRY EXPERIENCE

American Express · Gurugram, India *Aug 2024–Jan 2026*
Engineer II, Next Generation Data Management

- Built an LLM-based data-architecture documentation system that reduced manual documentation effort by approximately 60% and contributed to an early promotion to Engineer II.
- Developed Python REST APIs for infrastructure provisioning and worked with enterprise-scale distributed data systems.

IDFC FIRST Bank · Mumbai, India *Feb 2024–Jun 2024*
Data Science Intern, New Age Special Projects

- Fine-tuned BERT and open-source language models for real-time transaction classification, reaching approximately 99% accuracy on an internal evaluation set.
- Built PySpark and AWS pipelines for ingesting, preprocessing, and analysing large-scale transaction records.

Tally Solutions · Bengaluru, India *May 2023–Jul 2023*
Engineering Intern, CTO's Office

- Designed Go chaincode and REST APIs for a Hyperledger Fabric platform supporting MSME credit workflows; containerized peers, orderers, certificate authorities, and application services.

LEADERSHIP AND SERVICE

Technical Society Secretary, NIT Rourkela – elected; ran the technical events at Innovision 2023–2024
Coordinator, Career Development Centre, NIT Rourkela – trained students in software and analytics 2023–2024
Chief Coordinator, editorial board of the campus student media body – led the rebuild of its website 2021–2022
Core Organizer, HackNITR 3.0 – one of India's largest student-run hackathons (5,000+ participants) 2021

HONORS AND AWARDS

2026 ACM SIGMOD/PODS India Fellowship Recipient
2025 American Express Early Careers Hackathon – Synthetic data generation with GANs Winner
2023 IDFC FIRST Bank FAME 2.0 – EMI default prediction and collections modelling Runner-up
2023 Flipkart GRID 5.0 – Diabetic-retinopathy detection using quantum computing Semi-finalist
2023 Unstop UG Leaders national competition All-India Rank 4
2022 Tally CodeBrewers – Systems programming for file tagging Runner-up

TECHNICAL SKILLS

Quantum software: Qiskit, PennyLane, TKET, Cirq, Braket; circuit transpilation, noisy simulation, resource estimation

Systems and HPC: Python, Slurm, Parsl, PySpark, SQL, REST APIs, Git, Linux, AWS, Docker, Kubernetes

Machine learning: PyTorch, TensorFlow/Keras, graph neural networks, LLM fine-tuning, retrieval-augmented generation, GANs